

## Multiple Choice (10 marks)

1. Which of the following is true of the assessment of power in the arms?
  - (a) Extension at the elbow is produced by the biceps muscle
  - (b) Grade 3/5 (MRC scale) represents movement of the arm only when gravity is eliminated
  - (c) Grade 2/5 power represents movement of the arm only against gravity.
  - (d) The brachioradialis muscle is responsible for shoulder abduction
  - (e) Grade 0/5 power represents full range of motion with resistance
2. The influence of genetics on longevity is estimated to be about
  - (a) 33%
  - (b) 40%
  - (c) 60%
  - (d) 20%
  - (e) 80%
3. The World Health Organization recommends zinc supplements for which groups of people?
  - (a) Men with high physical activity levels
  - (b) All women over the age of 50
  - (c) All children in high income areas with low prevalence of stunting
  - (d) Elderly people with low incomes
  - (e) Adolescents in high-stress environments
4. The coronal suture joins the
  - (a) parietal, squamous temporal and greater wing of the sphenoid.
  - (b) left and right parietal bones.
  - (c) parietal and sphenoid bones.
  - (d) frontal and occipital bones.
  - (e) squamous temporal and greater wing of the sphenoid.
5. Which of the following appears to lower bad cholesterol?
  - (a) Vitamin E
  - (b) Vitamin A
  - (c) Thiamine
  - (d) Niacin
  - (e) Vitamin B12

## True / False (10 marks)

*Answer True or False*

6. True or False: Parvoviruses only replicate in dividing cells, which limits their ability to infect non-d
7. True or False: Among the various factors studied, behavioral factors have received significant res
8. True or False: Enzyme assays are primarily used to diagnose carriers of Huntington's disease ba
9. True or False: People with no previous history of mental health issues are more likely to experien
10. True or False: The cardiac cycle includes the phases of systole, diastole, and rest, which are criti

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the implications of a relative risk ratio of 1.2 for hip fractures in elderly women who exerc health outcomes.
12. Discuss the implications of a mutation in the repressor of the lac operon in E. coli that impedes its binding to lactose, particularly in the context of low glucose levels and gene expression.

*End of Paper*